Ch 2.2.3: Intro to classification

Lecture 9 - CMSE 381

Michigan State University
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Dept of Computational Mathematics, Science & Engineering

Mon, Feb 2, 2026

Announcements

11	F	2/6	Multiple Logistic Regression / Multinomial Logistic Regression	4.3.4-5	HW #2 Due Sun 2/8	
	M	2/9	***Project Day & Review***			
	W	2/11	**Midterm #1**			
12	F	2/13	Leave one out CV	5.1.1, 5.1.2		
13	M	2/16	k-fold CV	5.1.3		
14	W	2/18	More k-fold CV	5.1.4-5		Q5
15	F	2/20	k-fold CV for classification	5.1.5		
16	M	2/23	Subset selection	6.1		
17	W	2/25	Shrinkage: Ridge	6.2.1		
18	F	2/27	Shrinkage: Lasso	6.2.2	HW #3 Due Sun 3/1	
	M	3/2	Spring Break			
	W	3/4	Spring Break			
	F	3/6	Spring Break			
19	M	3/9	PCA	6.3		
20	W	3/11	PCR	6.3		Q6

Last Time:

- Finished Linear Regression

Announcements:

- Homework #2 Due Sunday Feb 8
- Next Monday - Review day
  - depends on what you ask!
- Wed 2/11 - Exam #1
  - Bring 8.5x11 sheet of paper
  - **Handwritten** both sides
  - Anything you want on it, but must be your work
  - Must have your name and group number
  - You must turn it in

Covered in this lecture

- Ch 2.2.3
- Error rate (classification)
- Bayes Classifier
- K-NN classification

Section 1

Classification Overview

What is classification

Classification: When the response variable is qualitative

- qualitative: Unordered set
- Examples
  - Eye color $\in \{brown, blue, green\}$
  - $email \in \{spam, notspam\}$

- Given feature vector X and qualitative response Y in the set S, the goal is to find a function (classifier) $C(X)$ taking X as input and predicting its value for Y.
- We are more interested in estimating the probabilities that X belongs to each category

Some examples

Discuss what set Y comes from in each case

- Predict whether a COVID19 vaccine will work on a patient given patient's age
- An online banking service wants to determine whether a transaction being performed is fraudulent on the basis of the user's IP address, past transactions, etc.

Section 2

Ch 2.2.3: Classification

Error rate

- Training data:
  $\{(x_1, y_1), \cdots, (x_n, y_n)\}$ with y_i qualitative
- Estimate $\hat{y} = \hat{f}(x)$
- Indicator variable
  $\mathrm{I}(y_i \neq \hat{y}_i)$ *is 1 if not the same (misclassified) and 0 else*

Training error rate:

$$\frac{1}{n}\sum_{i=1}^{n} \mathrm{I}(y_i \neq \hat{y}_i)$$

Test error rate:

$$\mathrm{Ave}(\mathrm{I}(y_0 \neq \hat{y}_0))$$

Good classifier: one that minimizes test error rate

Best ever classifier

We can't have nice things

Bayes Classifier:
Give every observation the highest probability class given its predictor variables

$$\Pr(Y = j \mid X = x_0)$$

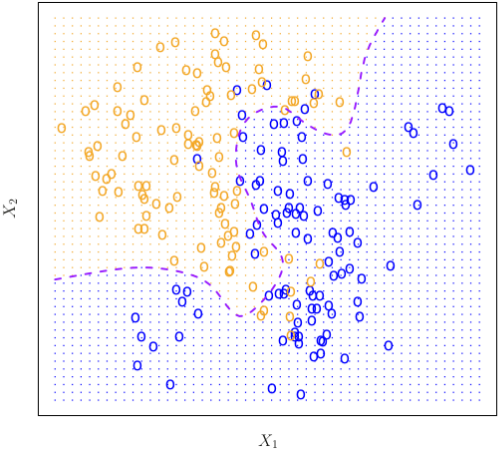
- It is possible to show (though the proof is outside of the scope of this book) that the test error rate is minimized, on average, by a very simple classifier that assigns each observation to the most likely class, given its predictor values.

An example

- Survey students for amount of programming experience, and current GPA
- Try to predict if they will pass CMSE 381.
- If we have a survey of all students that could ever exist, we can determine the probability of failure given combo of those features.

- No single classifier will get it right 100% since students with the same features might pass or fail

Bayes decision boundary



- Example where we simulated the data, so we know the conditional probability of each value of X_1 and X_2, using Bayes Theorem:

$$P(Blue|X_1, X_2) = \frac{P(X_1, X_2|Blue)P(Blue)}{P(X_1, X_2)}$$

$$Posterior = \frac{likelihood \cdot prior}{marginal}$$

- The purple line is where we switch our predictor, called the Bayes decision boundary

Bayes error rate

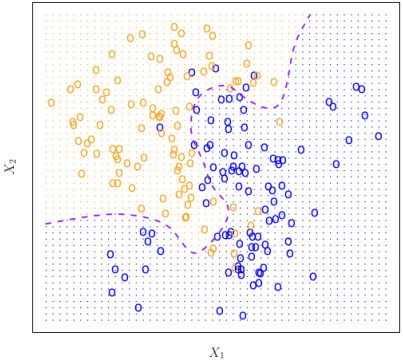
- Error at $X = x_0$ *The j with the highest value is what's predicted by Bayes for that x_0, so this is probability that the Bayes classifier is wrong*

$$1 - \max_j \Pr(Y = j \mid X = x_0)$$

- Overall Bayes error: *This is expectation over all x_0. This is the analagous to irreduceable error*

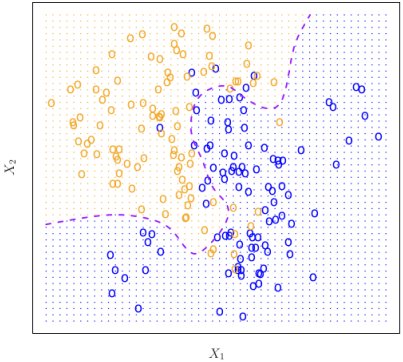
$$1 - E\left(\max_j \Pr(Y = j \mid X = x_0)\right)$$

This is the irreduceable error



The game

- Challenge: typically don't know the ground truth likelihood
- Game: do as good a job as possible estimating the Bayes decision boundary, by guessing the posterior $P(Blue|X_1, X_2)$ and $P(Yellow|X_1, X_2)$
- Classify given observation with highest estimated probability

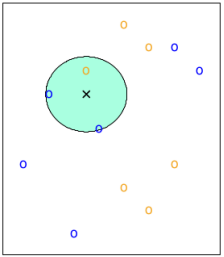


Test your understanding: PollEv

Section 3

K-Nearest Neighbors Classifier

K-Nearest Neighbors

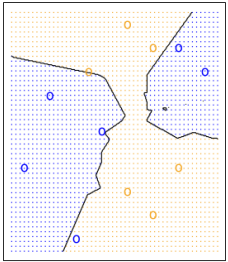


$K = 3$

- Fix K positive integer
- $N(x)$ = the set of K closest neighbors to x
- Estimate conditional proability

$$\Pr(Y = j \mid X = x_0) = \frac{1}{K} \sum_{i \in N(x_0)} \mathrm{I}(y_i = j)$$

- Pick j with highest value

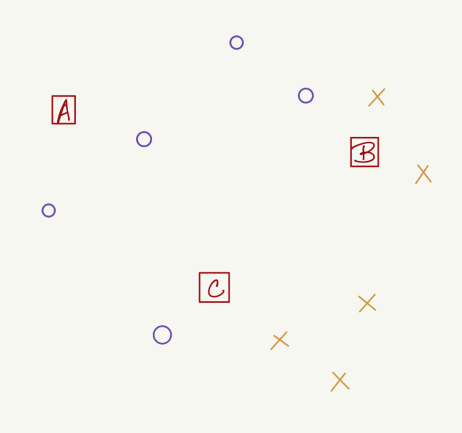


Black line: KNN decision boundary

Note this is not Bayes like in 2 slides! stupid textbook

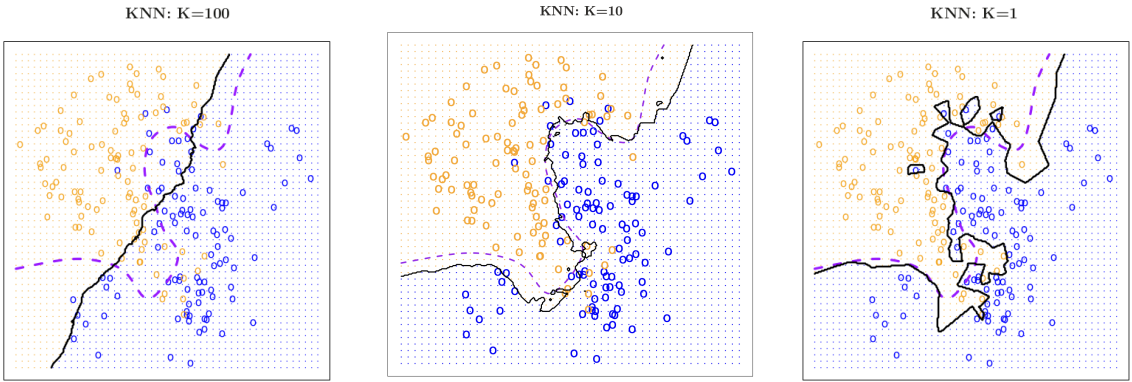
Example

Here label is shown by O vs X. What are the knn predictions for points A, B and C for $k = 1$ or $k = 3$? *Next year, maybe make an example like this with more than two levels, and/or try to decide what locations get which predictions for k=1 in prep for the next slide*



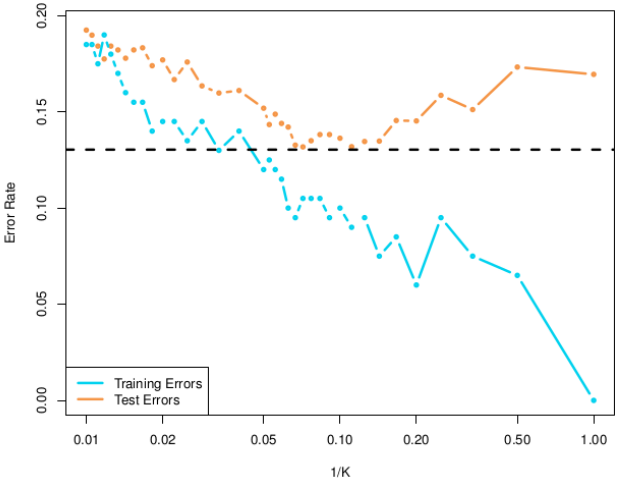
Point	$k = 1$ Prediction	$k = 3$ Prediction
A	*O*	*O*
B	*X*	*X*
C	*O*	*X*

Tradeoff



- Purple is Bayes decision boundary. Same on both
- $K = 1$, boundary is overly flexible, finds patterns not there
- $K = 100$ not enough flexibility

More on tradeoff



- right side, $K = 1$, training is 0 error but high test error
- Test error has same U shape as bias-variance tradeoff in regression setting: decline at first, then increase again when overfitting

Jupyter notebook

Next time

CMSE381_S2026_Schedule : Sheet1

Lec #	Date		Topic	Reading	HW	Pop Quizzes	Notes
1	M	1/12	Intro / Python Review	1			
2	W	1/14	What is statistical learning	2.1		Q1	
3	F	1/16	Assessing Model Accuracy	2.2.1, 2.2.2			
	M	1/19	MLK - No Class				
4	W	1/21	Linear Regression	3.1		Q2	
5	F	1/23	More Linear Regression	3.1	HW #1 Due Sun 1/25		
6	M	1/26	Multi-linear Regression	3.2			
7	W	1/28	Probably More Linear Regression	3.3		Q3	
8	F	1/30	Last of the Linear Regression				
9	M	2/2	Intro to classification, Bayes classifier, KNN classifier	2.2.3			
10	W	2/4	Logistic Regression	4.1, 4.2, 4.3.1-3		Q4	